Bayesian Optimisation fo Discrete Event Simulation Experiments

Harry Newton

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1 Experiment: Comp_01

In this experiment we compare the performance between three configurations:

- *Vanilla*: SK-HGP GP with Expected Improvement
- *AEI*: SK-HGP with Augmented Expected Improvement
- *IG* SK-HGP with Information Gain

All three experiments began with k equally spaced intial values. Each was sampled n times to produce a mean evaluation y with variance σ_ε. Each data point can be summarised by the set $(x, n, y, \sigma_\varepsilon)$. Each experiment was run over M macroreplications and each BO instance was run over T iterations. The table bellow shows the values used for this experiment:

Table 1: Experimental paramaters.

Paramater	Value
k	5
M	25
T	10
n	10
seed	12345

1.1 Test Function

The test function for this experiment is defined as:

$$f(x) = \sin(5x) + \cos(7x).$$

And the noise function is defined as:

$$\sigma_\varepsilon^2(x) = (0.3 + \tau \exp(-0.5((x - \phi)/0.1)^2))^2$$

The paramatarisation of the noise function, the x and n domains can be found in the table below.

Table 2: Test Problem paramater values and domain. Replication domain is also included

Paramater	Value
x	$[0, 1]$
n	$[3, 100]$
ϕ	0.6
τ	0.7

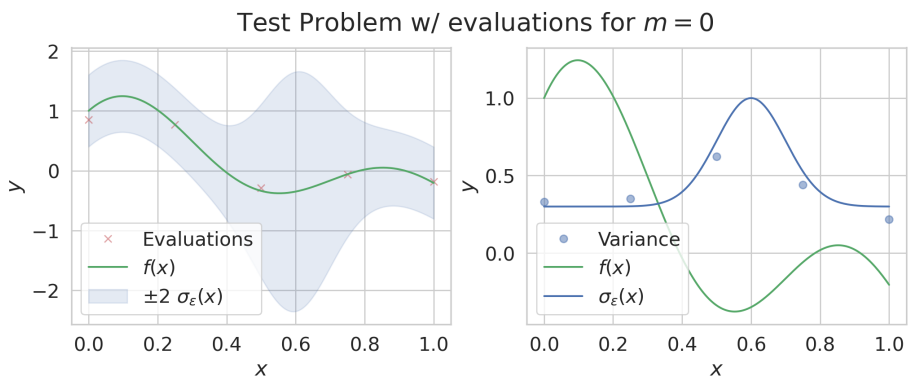

Figure 1: Test problem to be optimised with the true signal and noise functions present. Initial evaluations for $m = 0$ are also present

1.2 Results:

The results are below for the most recent experiment. Here f^* is the cumulative best predicted optimal solution from the SK-HGP model at a given n. Previously, I would have fed the predicted maximisers $\hat{x}^*(n)$ into the test function and calculated that instead to reduce the uncertainty from the predicted optimal values.

Both AEI and IG do better than vanilla but there is naturally a lot more variability in their traces. I think maybe an error bar plot over average cumulative n and average cumulative f^* might be visually clearer.

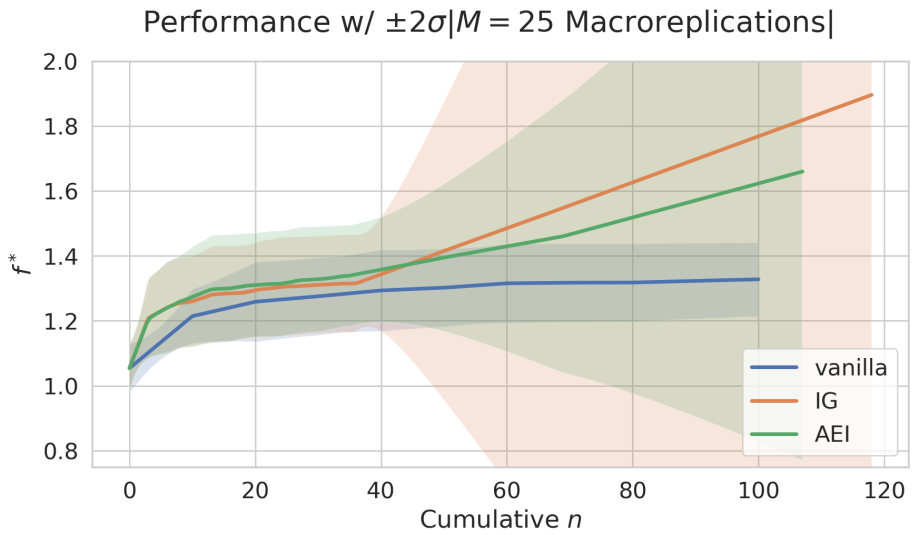

Figure 2: Performance Comparison between models across 25 macroreplications.